Lemma 1 - Fast (Var

(Algorithm 1 computes q^+)

Note 1: I am assuming the algorithm is in while loop form, although proof can be adopted for recursive form.

Note 2: Assuming that no duplicate elements are present in the array for now.

General structure for proof:

First I prove that q^+ is a specific outcome of the random variable for $0 \leq \alpha < 1$. Then I show that at every iteration of the algorithm, the element corresponding to q^+ remains in the array & Thus at termination, when only on element remains this is the q^+.

Detailed proof on next page

Step 1: Prove that q^+ is a specific outcome of the random variable X.

First the definition of $q^+(x)$:

$$q^+(x) = \max\{\tau \in \bar{\mathbb{R}} \mid P[X < \tau] \leq \alpha\}$$

We start by observing the function $f(\tau) = P[X < \tau]$. This is the CDF but with a strict inequality. We assume the random variable X has n outcomes, where $\tau_1 < \tau_2 \ldots < \tau_n$.

The function $f(\tau)$ takes n discrete values, where each unique values of the function other than 0 and 1 exists for the domain $(\tau_k, \tau_{k+1}]$. For function values 0 and 1, the domain is $(-\infty, \tau_1]$ and (τ_n, ∞).

As we can see, for every α such that $0 \leq \alpha < 1$, the maximum τ such that the inequality is satisfied is one of τ_1 through τ_n. Thus the value of $q^+_\alpha(x)$ for all admissible α is one of the potential outcome of discrete random variable X.

Step 2: From step 1 of the algorithm, we can prove that at the beggining (before any loops), the array x contains $q_a^+(x)$ for all admissible a. We now show the loop invariant is the $q_a^+(x)$ always exists in the array x.

In every iteration 3 cases happen:

Case 1: Tail is selected such that $a \leq$ tail. In this case, $P[x < x[pivot]]$ is greater than or equal to a. Thus $q_a^+(x)$ is included in the set of values from x[0] to x[pivot]. The only values removed from x are those greater than x[pivot], thus loop invariant that $q_a^+(x)$ remains in x is held.

Case 2: Tail is selected such that $a >$ tail. In this case $P[x < x[pivot]]$ is less than a. Thus $q_a^+(x)$ is included in the set of values for greater than x[pivot], which are the only values that are not removed. Thus $q_a^+(x)$ remains in x after iteration.

TO DO: Prove $CVAR_{a-tail}(new\ x) = CVAR_a(old\ x)$

Case 3: The array x only has one element:
If the original array x contains one element, probality of this outcome is 1. Thus $q_a^+(x)$ = equals this element for all admissible a.
If the original array x contained more than one element, all the elements that were removed could not have been $q_a^+(x)$, and since $q_a^+(x)$ has to be in original array x, this element is the solution.

Therefore, at termination one element will remain which is $q_a^+(x)$.